Predictive Modelling to Predict Students Dropout and Academic Success

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Recap: Project Motivation and Research Questions

WHY THIS TOPIC?

Student dropout and academic failure have high costs for students and institutions:

- lost tuition revenue
- delayed career progress
- lower graduation and retention rates

Using data-inclined approaches, Universities can:

- Boost Graduation Rates
- Optimize Resource Allocations
- Develop Evidence-Based Policies

KEY STUDY QUESTIONS

1. What are the biggest contributors to student dropout and academic failure?
2. Can we predict whether a student will drop out, remain enrolled, or graduate?
3. Can we predict dropout using only first-semester data?

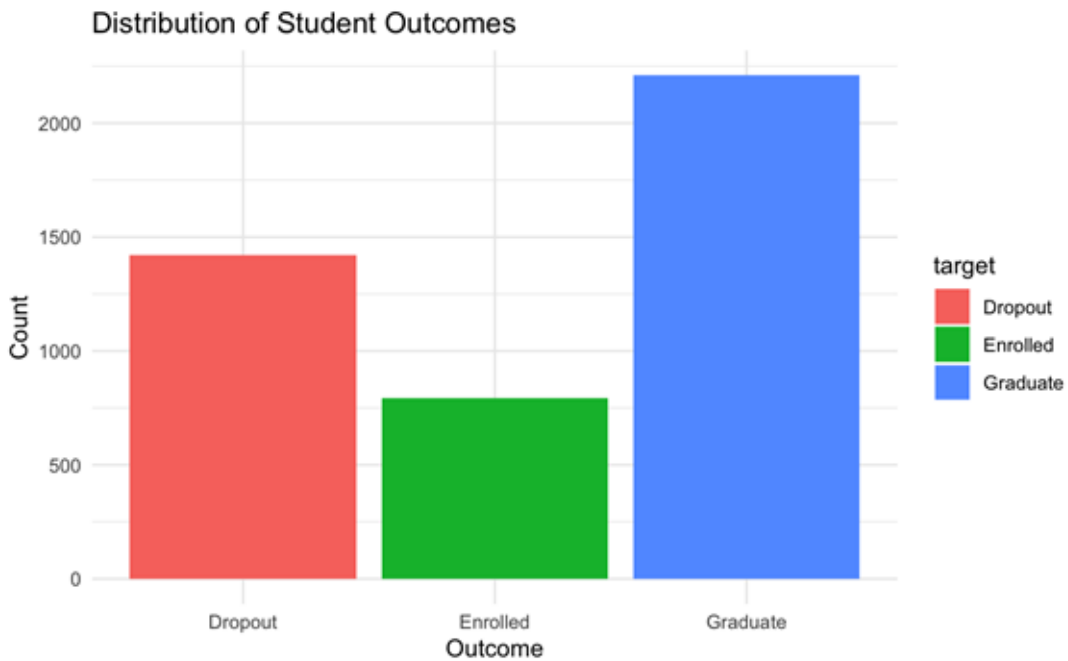
Recap: Data Overview

Polytechnic Institute of Portalegre, Portugal | 2008-2019| **4,424** Student Records| **37** features

Features in the data include:

1. Marital status
2. Application order
3. Daytime/ evening attendance
4. Gender
5. Tuition fees up to date
6. Scholarship holder
7. International
8. Debtor
9. Curricular units 1st sem (enrolled)
10. Curricular units 1st sem (credited)
11. Inflation rate
12. etc

Our dependent variable is categorical with 3 classes (**Dropout, Enrolled, Graduate**)



Methodology Note: Data was split 70/30 for training and testing on all models.

Data source: UCI Machine Learning Library

Logistic Regression: Data Modelling

- Dependent variable: ‘target’
- Independent variable: All other variables remaining (36)
- The dependent variable is of categorical type with 3 classes: ‘ Dropout’, ‘Graduate’, and ‘Enrolled’.
- We developed a Multinomial Logistic Regression - our reference level in the dependent variable was ‘Graduate’

Data Splitting (70:30)			**Data Size**
Graduate	Dropout	Enrolled	Training Data
1546	994	555	3095
Graduate	Dropout	Enrolled	Testing Data
663	427	239	1329

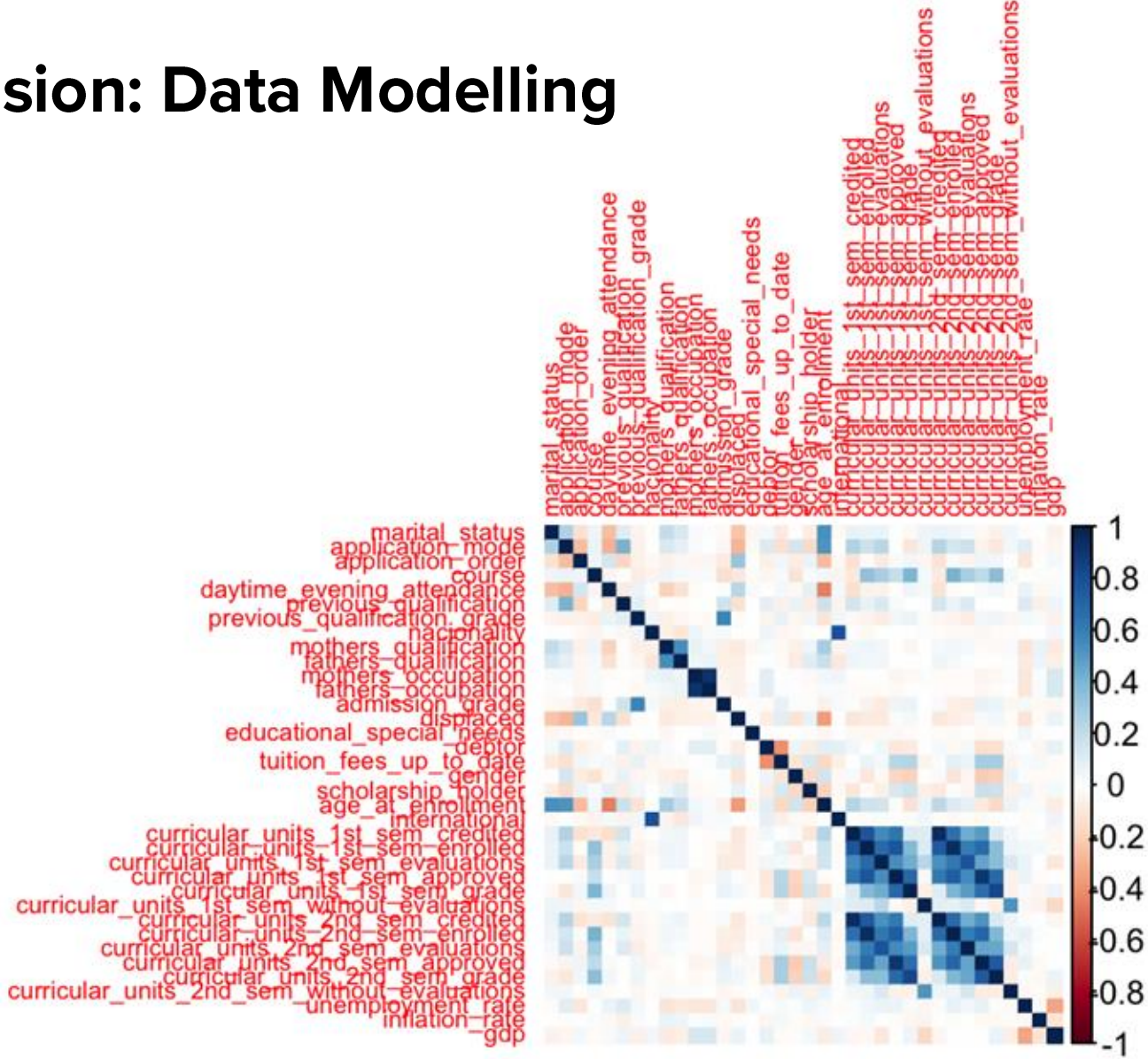
Class balance was preserved in both data sets		
Graduate	Dropout	Enrolled
0.4995153	0.3211632	0.1793215
Graduate	Dropout	Enrolled
0.4988713	0.3212942	0.1798345

We checked for collinearity using the Variable inflation factor and removed more than 10 variables, some of which include:

- Mother’s occupation
- Father’s qualification
- Curriucular units 1st sem credited
- Curriucular units 1st sem enrolled
- Curriucular units 1st sem approved
- Curriucular units 1st sem grade
- Curriucular units 2nd sem credited
- Curriucular units 2nd sem enrolled

Logistic Regression: Data Modelling

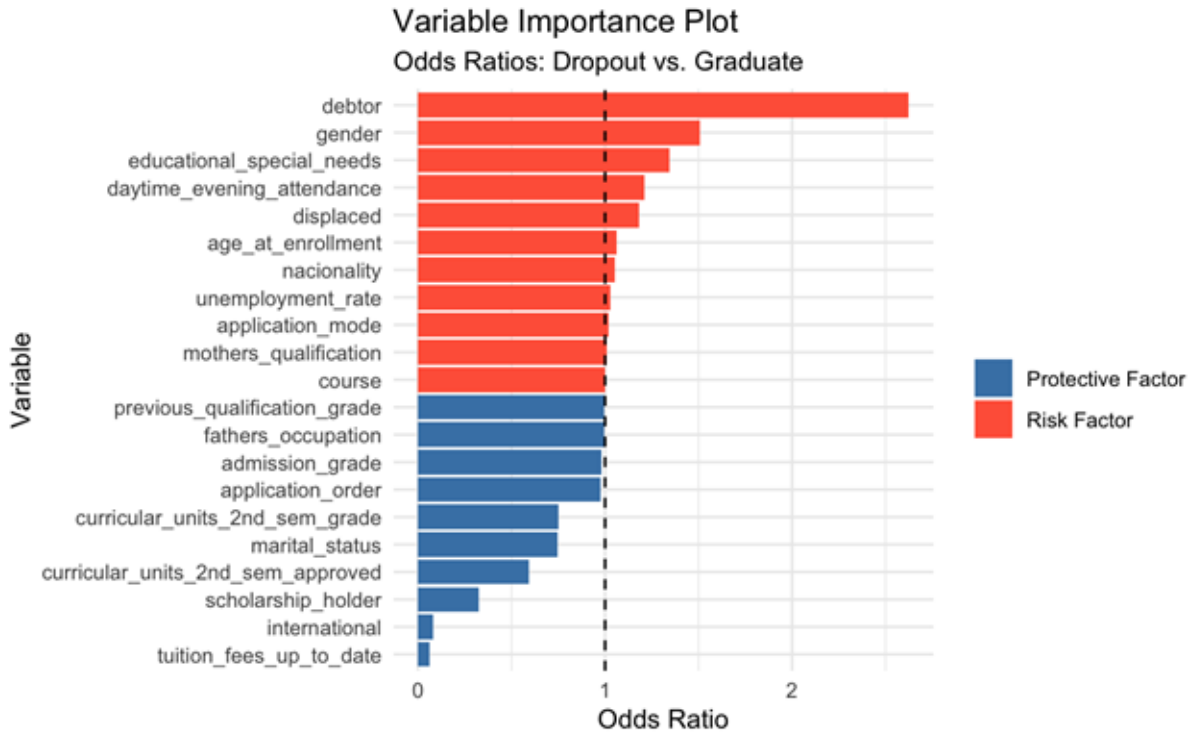
- Variables with a VIF of more than 10 were removed (very high correlation)
- We did a pairwise correlation for variables with a VIF with more than 5 but less than 10 (moderate correlation and pick the variable that was more meaningful.



Logistic Regression: Model Results

AIC	4101.955	This is a good model, 36% of the variation in the data can be explained by the model. A McFadden's Pseudo R squared of between 0.2 - 0.4 is considered a good fit for logistic regression.
McFadden Pseudo R-squared	0.3641	

Which predictors matter most?



Everything to the right of the dashed line (OR>1) = increases dropout risk compared to graduating and

- Our strongest risk factor to dropout is debt.
  - Being a debtor makes a student **2.6x more** likely to dropout than graduate.
- Having **tuition fees up to date** (financial stability) is the strongest positive factor.
  - Students with fees paid are **94% less** likely to drop out.
- Financial factors dominate both sides.
  - Financial support such as education bursaries, scholarships, etc are the single most powerful intervention point.
- Academic performance in the 2nd semester (units approved) is the **second most important domain**.
- **Demographics** like gender, age, special needs play a role but are **less dominant than financial and academic factors.**

Logistic Regression: Model Evaluation

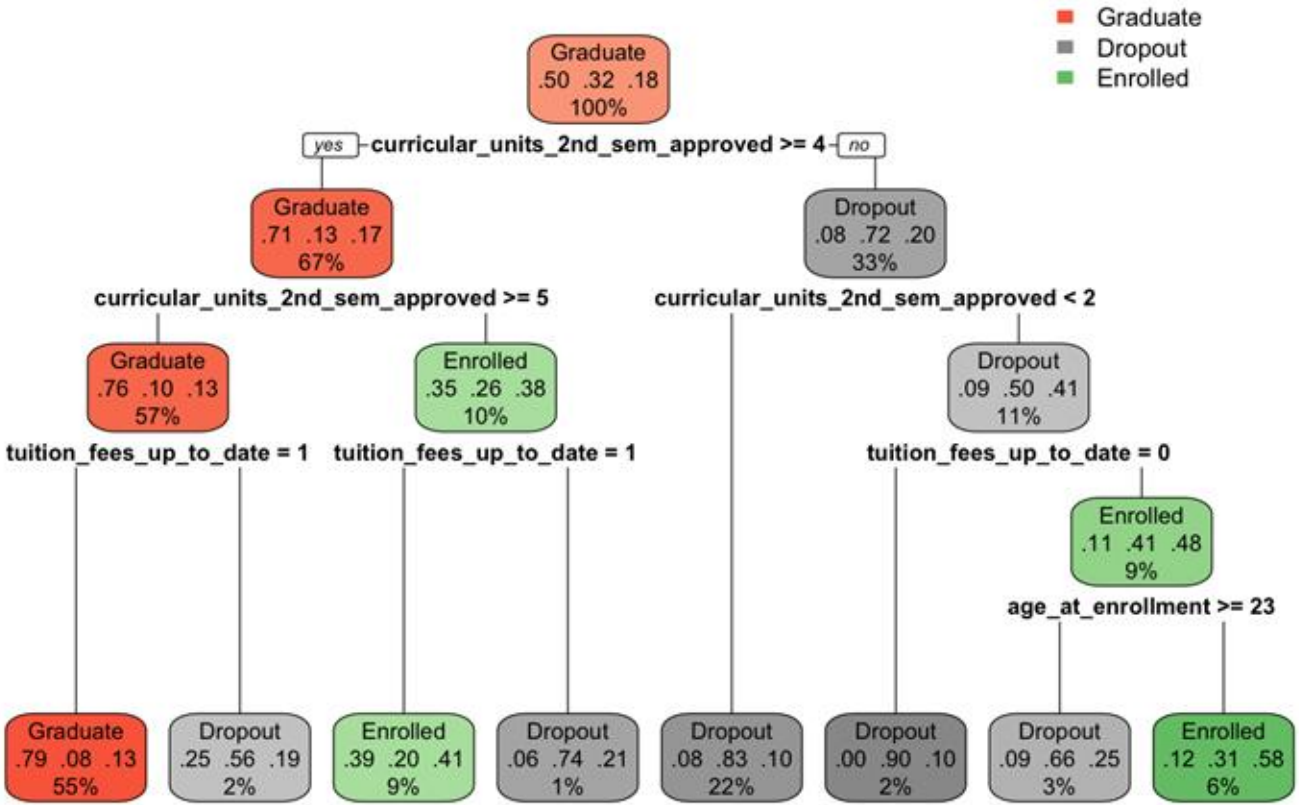
Model Size			
	Graduate	Dropout	Enrolled
Testing data	663	427	239

Confusion Matrix			
Prediction	Graduate	Dropout	Enrolled
Graduate	613	69	**126**
Dropout	28	326	**63**
Enrolled	22	32	50

Model Accuracy	**74.40%**		
Statistics by Class			
	Graduate	Dropout	Enrolled
Sensitivity	**92.40%**	**76.40%**	**20.90%**
Balanced Accuracy	**81.50%**	**83.30%**	**57.90%**

The model is struggling to capture the ‘Enrolled Class’.

Classification Tree: Model Results



- Most important variable: **"Curricular_units_2nd_sem_approved"**.
- Financial factor is still critical, '**tuition_fees_up_to_date**' appears on both branches. Even good students drop out without it.
- **The tree confirms our logistic regression model** since both models agree on the same top predictors i.e., curriucular_units_2nd_sem_approved (strongest positive factor), tuition_fees_up_to_date (strongest protective factor), age_at_enrollment(mild risk factor).

Classification Tree: Model Evaluation

Model Size			
	Graduate	Dropout	Enrolled
Testing data	663	427	239

Confusion Matrix			
Prediction	Graduate	Dropout	Enrolled
Graduate	590	52	95
Dropout	32	324	55
Enrolled	41	51	89

Model Accuracy	75.47%		
Statistics by Class			
	Graduate	Dropout	Enrolled
Sensitivity	88.99%	75.88%	**37.23%**
Balanced Accuracy	83.46%	83.12%	**64.39%**

- The model struggles to capture students who fall in the 'Enrolled class' (**37.2%**)
- This is most likely because the 'Enrolled Class' is a middle ground category, so its features overlap with both other categories.
- Comparing the logistic regression model to the classification model, the classification model wins! The classification model has a higher accuracy (**75.5% vs 74.4%**), and higher enrolled balanced accuracy (**64.4% vs 57.9%**).

Random Forest: Model Results

Model Size			
	Graduate	Dropout	Enrolled
Testing data	663	427	239
Training data	1546	994	555

Confusion Matrix				
Prediction	Graduate	Dropout	Enrolled	Class Error
Graduate	1437	50	**59**	**0.07050453**
Dropout	161	745	88	0.25050302
Enrolled	**256**	**151**	148	**0.73333333**

OOB Error Rate	24.72%
Accuracy	75.28%

- High error rate stands out.
- While the model is great at identifying students who graduate, it is **struggling significantly** to distinguish between students who are have Dropped out and those who have Graduated.
- Our accuracy is still good but **lower than the classification model** (75.28% vs 75.47%); however, it is **better than the logistic regression model** (74.40%)

Check for Class Imbalance		
Graduate	Dropout	Enrolled
1546	994	**555**

- Class imbalance confirmed.
- We have nearly three times as many Graduates as Enrolled students.
- The model is biased toward the "Graduate" class because it has the most observations (1,546). The "Enrolled" class is the smallest, leading the model to misclassify 256 Enrolled students as Graduates!

Balanced Random Forest: Model Results

- Goal: To sample an equal number of rows from each class when building the tree.
- Reference sample size: 555 (smallest class size)

Model Size			
	Graduate	Dropout	Enrolled
Testing data	663	427	239
Training data	**555**	**555**	**555**

Confusion Matrix				
Prediction	Graduate	Dropout	Enrolled	Class Error
Graduate	1309	55	**182**	**0.1533**
Dropout	119	711	164	0.2847
Enrolled	**181**	**102**	**272**	**0.5099**

OOB Error Rate	25.95%
Accuracy	74.05%

- **Enrolled vs. Graduate:** 182 Graduates were mistaken for Enrolled, and 181 Enrolled were mistaken for Graduates. These two classes are still bleeding into each other, likely because their academic data (grades/credits) looks very similar.
- **The Trade-off:** The **Graduate** error doubled (from ~7% to 15%). This is the standard "price" of balancing; the model is no longer biased toward the majority, so it makes more mistakes there to get the minority class right.

- Much better results!
- By downsampling, we cut the **Enrolled** error rate from **73% down to 51%**.
- Model accuracy dropped slightly (from 75.28% to 74.05%); however, the model is now significantly

Balanced Random Forest: Model Evaluation

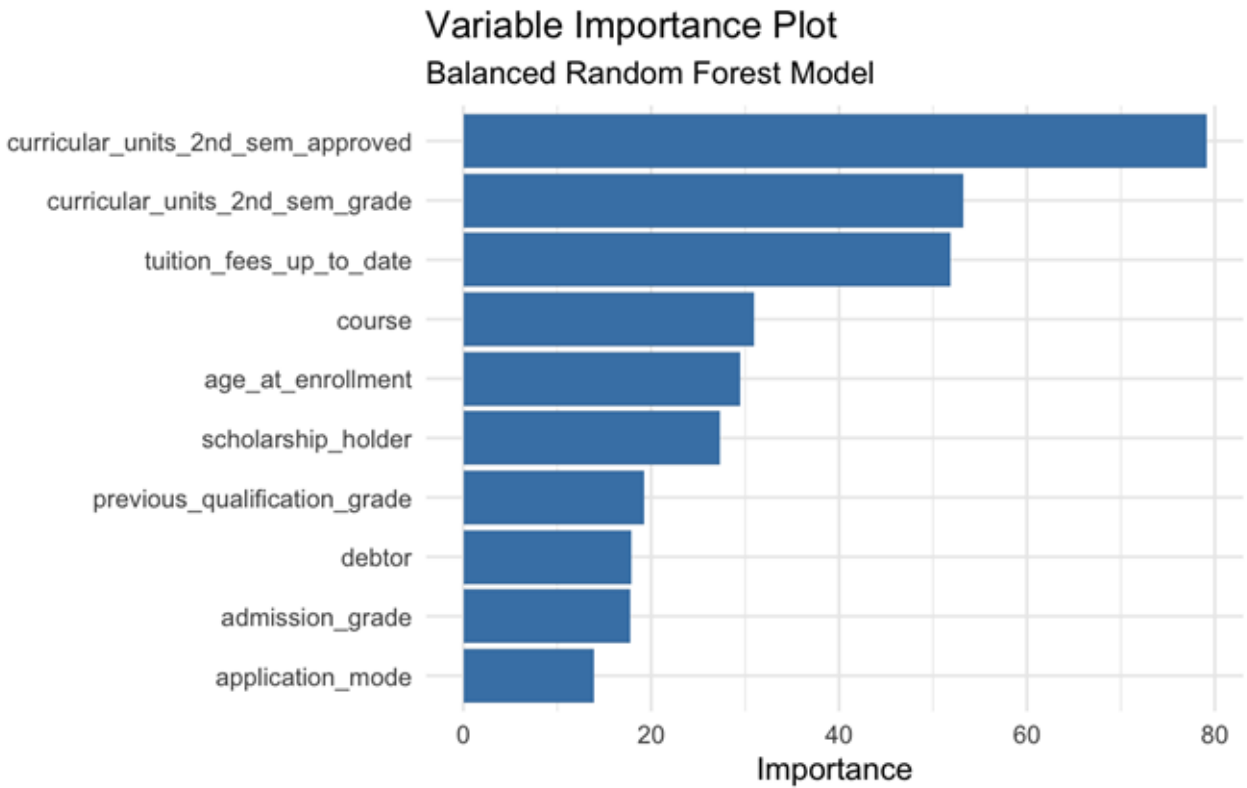
Model Size			
	Graduate	Dropout	Enrolled
Testing data	663	427	239

Confusion Matrix			
Prediction	Graduate	Dropout	Enrolled
Graduate	572	45	**73**
Dropout	20	321	38
Enrolled	**71**	**61**	128

Model Accuracy	**76.82%**		
Statistics by Class			
	Graduate	Dropout	Enrolled
Sensitivity	86.27%	75.18%	**53.56%**
Balanced Accuracy	84.28%	84.37%	**70.72%**

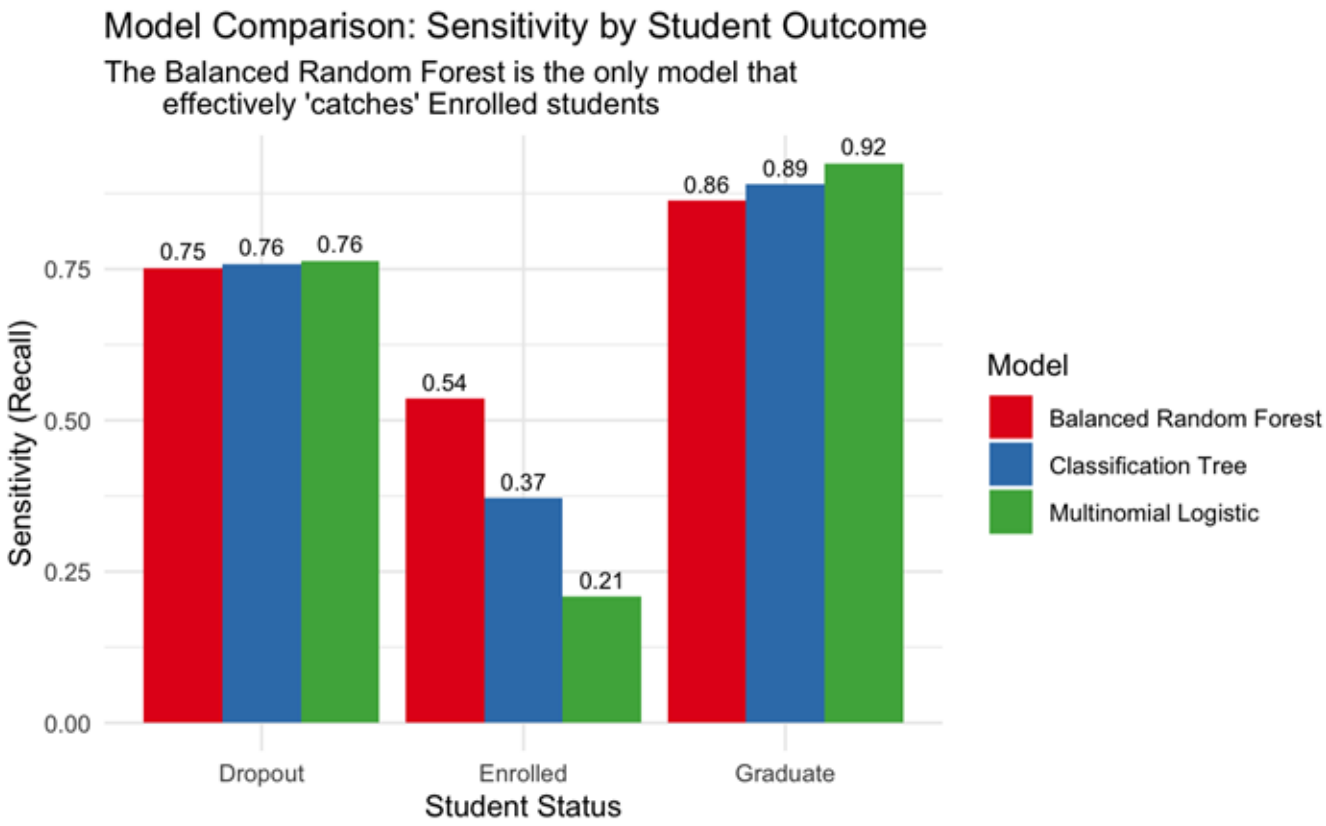
- The model is now able to capture more than 50% (53.56%) of the enrolled class; which it was struggling with previously, 37.23% in the classification model, and 20.90% in the logistic model.
- The model's accuracy is 77% (76.8%) which is the **highest of all the models** we've built, i.e.,Unbalanced random forest model (75.28%), logistic regression (74.4%), classification tree (75.5%).

Balanced Random Forest: Model Evaluation



- 'curricular_units_2nd_sem_approved' is the **most important variable** in predicting outcomes for our model.
- This confirms our finding from the classification tree.
- Tuition is the most important non-academic variable.
- Variables such as 'international', 'educational_special_needs', and 'application_order' have near zero importance.

Conclusion: Model Comparison



- The balanced Random Forest is our most fair and useful model as it provides the best 'safety net' for the Enrolled students without sacrificing too much performance on Graduates and Dropouts.
- The logistic regression remains the most accurate overall.